

OLIST E-COMMERCE ANALYTICS

An Independent Analysis of the Brazilian Market

By combining customer, order, product, and payment data, we identified opportunities for growth and improvement.

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Methodology

The path from raw data to insight

1

Data & Merging

7 tables (customer, order, product, seller, payment, review) merged into a single table via shared identifiers.

2

Exploratory Analysis

Trends and anomalies were examined across product, payment, and seller axes.

3

Statistical Testing

4 hypothesis tests confirmed whether observations were real or due to chance ($\alpha = 0.05$).

4

Visualization & Recommendations

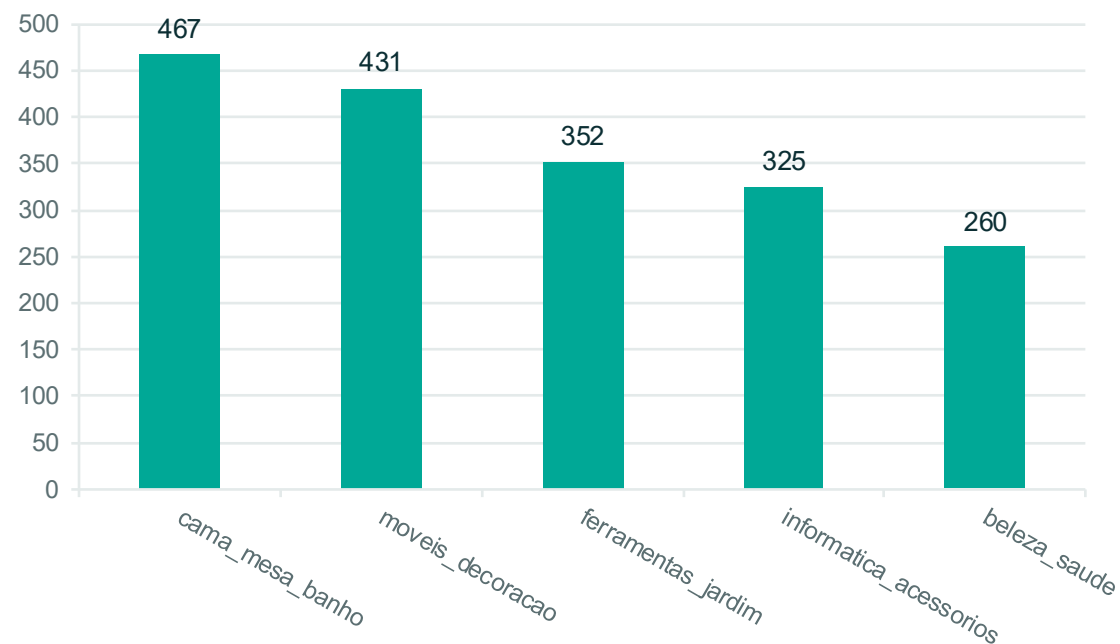
The strongest findings were visualized and translated into actionable recommendations.

Focus Questions: Which product/seller/region segments lead in volume, and which in revenue? What actually drives satisfaction?

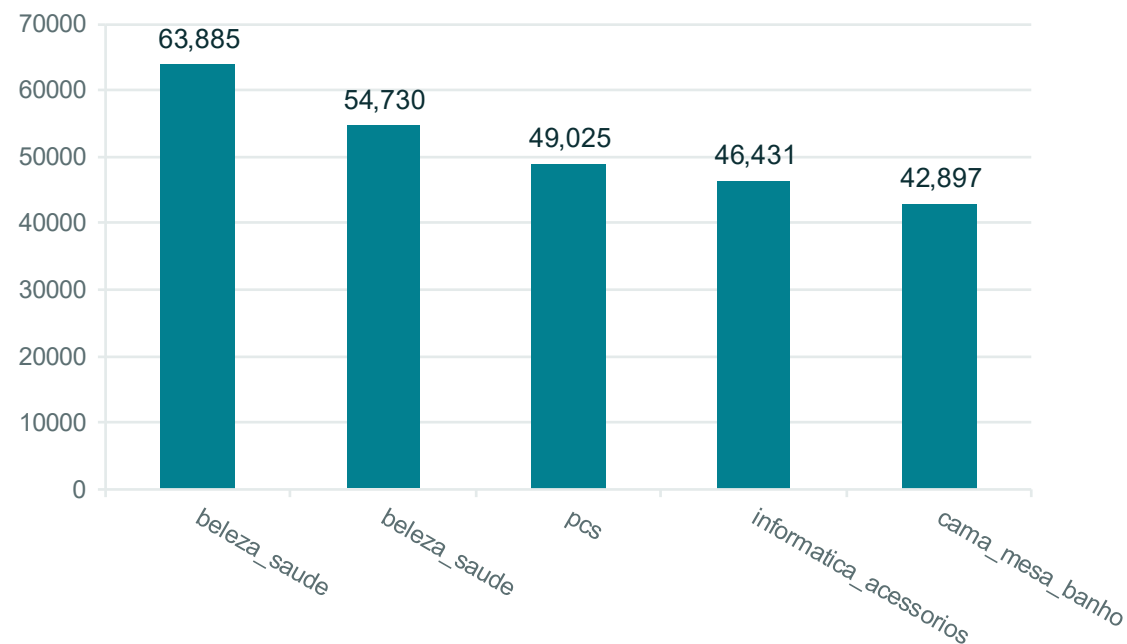
Product Segmentation: Volume ≠ Value

Best-selling products and highest-revenue products come from different categories

Top-Selling Categories (Volume)



Top-Revenue Categories (Value)



Opportunity: 618 products with at least 20 orders and a 4.7+ rating were identified — products that are both in demand and well-liked, yet absent from current commercial priorities. Increasing their visibility is a low-risk growth opportunity.

Strongest Finding: Delivery Time

As delivery time increases, customer satisfaction drops in a statistically significant way

$r = -0.30$

Moderate negative correlation

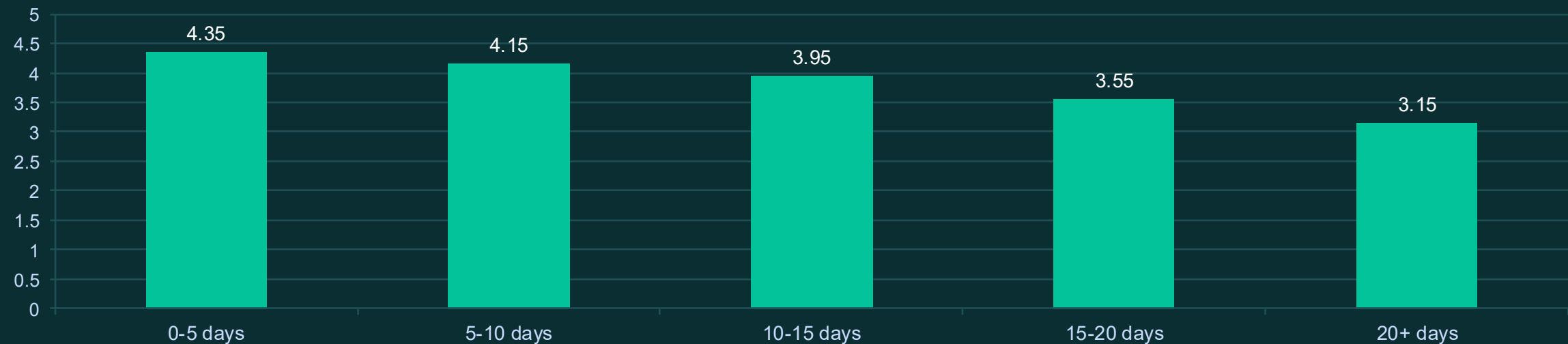
$p < 0.001$

Statistically significant

22 days

Slowest seller's avg. delivery

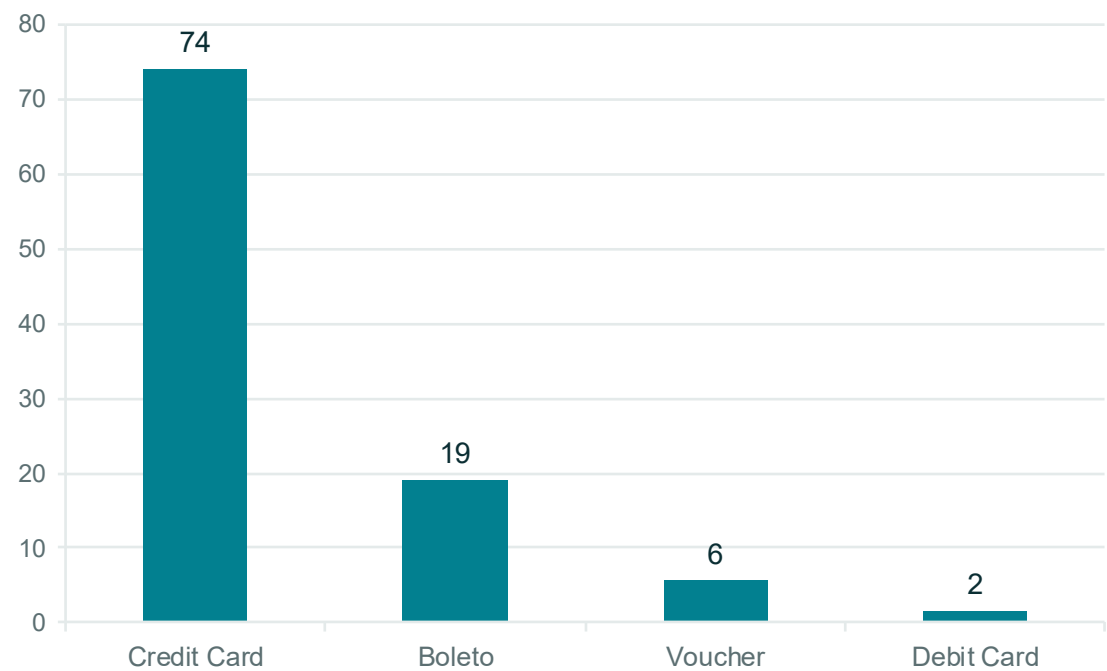
Average Review Score by Delivery Time Range



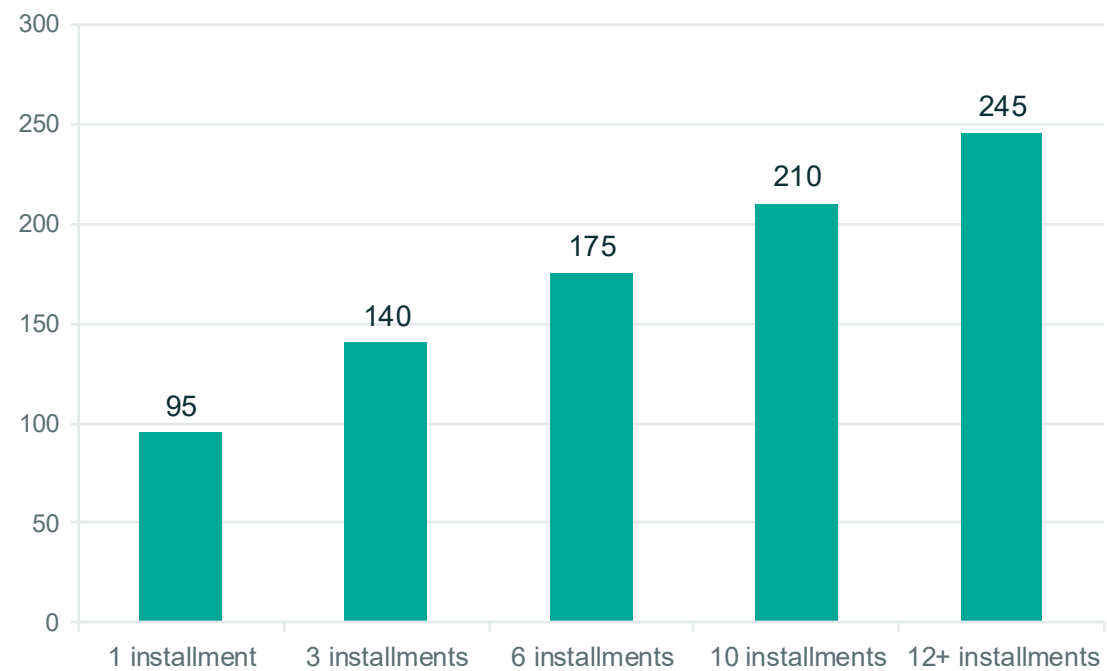
Payment Behavior

Credit card dominates; installment options increase spending amount

Payment Method Distribution (%)



Average Payment Amount by Installment Count



Finding: There is a significant, moderate positive relationship between installment count and payment amount ($r=0.27$, $p<0.001$). Payment method has no practical effect on satisfaction — installment options can be promoted without added risk.

Conclusion & Recommendations

Growth lies less in adding new products and more in making existing assets visible and improving logistics

1

Recurring Theme: Volume $\neq$ Value

The same pattern repeats across product, seller, and geography: SP has the highest order volume (41,746) but is among the lowest in average order value; GO reaches the highest average value with far lower volume (2,020). Marketing/inventory strategy should be built per segment.

2

Delivery Time — Priority Action

The strongest and most actionable finding ($r=-0.30$). Logistics improvement should be prioritized with sellers whose delivery times exceed 15 days.

3

Hidden Opportunity Products

618 products show high satisfaction and sufficient demand but low visibility. They should be promoted in recommendation engines and marketing.

4

Promoting the Installment Option

The installment option increases spending amount ($r=0.27$) without affecting satisfaction — it should be made more visible at checkout.

Detailed analysis, code, and the full dataset are available in the project notebook.